

ABSTRACT

The proposed review is to investigate **the interaction of demographic variables in LMS usage to explain achievement gaps**. Can demographic and LMS variables explain achievement gaps? Does prior education moderate the effect of LMS activity on grades? Are students from lower socioeconomic status (SES) backgrounds disadvantaged even with high LMS engagement? and Do age groups differ in how they translate LMS usage into performance? are a subset of questions to be investigated based on report specifications.

1. Related Work Contribution

Against the backdrop that distance learning faced high dropout rates and the need for motivation and engagement for retention, [Jawthari, M. et al, 2022]'s study examined how demographic characteristics (*gender, education, credits, etc.*) related to students' engagement in distance learning. The case study was for a 2014B Science & STEM module - 1,500 students. Engagement was measured using Virtual Learning Environment (VLE) activity and assessment behavior. A selected two-level K-Means Algorithm, an unsupervised clustering algorithm, using silhouette coefficient, successfully grouped students. Apriori Association Rules with *support* ≥ 0.1 , *confidence* ≥ 0.9 were used to identify relationships using metrics (*frequency, probability*) and students' engagement. Results revealed that *gender, highest education, credits, and previous attempts* positively correlated with engagement. Cited limitations: logs lacked session duration data and no reported penalty for late submission. Their findings were 1) engagement in distance learning can be identified using clickstream and submission metrics 2) a two-level engagement model worked best 3) demographics were strongly linked: *male, higher education, 60 credits, no prior attempts, and no disability* implied *higher engagement* (support=0.15, confidence=0.95) 4) *students with lower education, younger age, and 60 credits* were more likely disengaged, and 5) at-risks students were easily identifiable for intervention. Future work could include capture of time spend/clicks per session, exploration of the effects on grades, and **K-prototype algorithm** on mixed data.

The study by [Alshehri, A. et al, 2020] investigated how *Gender* and *Age* moderated students' acceptance and use of LMS in Saudi by hypothesizing: **H1: Gender moderates all relationships** and **H2: Age moderates all relationships**. The methodology used multi-stage cluster sampling and a three-part survey. Using the Unified Theory Acceptance And Use of Technology (UTAUT) model with 6 usability features, and data from 605 students, the authors applied **PLS-SEM** with multi-group analysis, a quantitative technique testing complex models to examine whether structural relationships differ across subgroups. The conclusion was that gender and age mattered only in how students translated facilitating conditions into actual LMS use. The analysis included separate Structural Models for both **Gender** and **Age**. Reliability and validity were confirmed using the **Composite Reliability** metric ($CR > 0.7$; $AVE > 0.5$). The findings: both gender and age moderated only one path—which implied Actual Use (AU), where females and younger students showed stronger perceptions. Implications for multiple policymakers were also realized. Discussions, consistent with prior studies, weighed on the fact that females relied more on facilitating conditions (*IT support, infrastructure*) than males. Also for the *Age* effect, younger students were more influenced by facilitating conditions - they depended more on institutional support for LMS use. Both moderators influenced only one path, implying growing familiarity with LMS across groups reduced demographic differences. As a result, universities were to strengthen IT infrastructure and support services, especially for females and younger students. The exploration of other demographics and longitudinal LMS trends are potential future research.

[Claar, C. et al, 2014] addressed 10 hypotheses, to *predict the positive relationships between TAM factors (PE, PU, AT, BI, AU) and demographic influences (age, race, gender, education)* “with willingness to change LMS as a proxy for acceptance”. The study investigated how demographic factors influenced students' acceptance of LMS. A TAM-based survey was administered and Regression Analysis was used to quantify and test the strength of relationships between TAM variables and demographics in predicting

LMS acceptance. The survey used a Likert Scale (7-point) of categorical questions. $N=429$ valid data points were collected by Survey Monkey. Both internal (social desirability bias, limited minority presence, etc.) and external (limited generalization, TAM-only basis, etc.) challenges were identified. Discussion and recommendations were of the nature: 1) TAM variables were strongly interrelated: ease of use $\rightarrow$ usefulness $\rightarrow$ attitude $\rightarrow$ intention, 2) unexpected finding: Behavioral Intentions (BI) did not lead to AU, and 3) demographic effects: a. Age $\rightarrow$ greater concern with usefulness, b. education $\rightarrow$ higher acceptance and ease of use, and c. Race and gender $\rightarrow$ no effect. The recommendation was for universities and LMS providers to emphasize *ease of use and usefulness* in marketing and orientation to enhance adoption. Future areas of study: larger/multi-university samples, comparisons across majors, colleges vs. universities, faculty/administrator perspectives, use of both TAM and UTAUT models, and Open-ended questions on important LMS features.

2. Description of Dataset

2.1. Overview of Dataset: The deciding dataset rested with the Open University Learning Analytics Dataset (OULAD) repository which housed 7 csv tables of which only 6 related to the discussion ***Student Performance and LMS Usages***. OULAD's Table descriptions follow:

Table	Description	Usages/Notes
StudentInfo	Student demographics + outcomes; Central dataset for identifying students Dim: $32,593 \times 12$; 3.3MB; Vars: <i>code_module, code_presentation, id_student, gender, region, highest_education, imd_band, age_band, prev_attempts, studied_credits, disability, final_result</i>	Demographic analysis, linking background factors to engagement and performance *Some missing imd_band; final_result $\in$ {Withdrawn, Fail, Pass, Distinction}
StudentRegistration	Info on registration status: registered, withdrew, or completed Dim: $32,593 \times 5$; 1.1MB; Type: Enrollment timeline; Vars: <i>code_module, code_presentation, id_student, date_registration, date_unregistration</i>	Studying dropout patterns (who withdrew and when); provides added insight *date_unregistration has ? values $\rightarrow$ convert to NA
studentAssessment	Assessment submissions (<i>assignment scores, submission date</i>) Dim: $173,912 \times 5$; 5.4MB; Type: Assessment-level scores; Vars: <i>id_assessment, id_student, date_submitted, is_banked, score</i>	Recommended for added depth: <i>studentAssessment + assessments $\rightarrow$ performance</i> on assignments/exams. vle $\rightarrow$ breakdown of LMS resource types. *Missing scores if not submitted; multiple per student
Assessments	Details of assessment structure (weight, deadline, type: exam, TMA, CMA, etc.) Dim: $10,065 \times 6$; Type: Assessment metadata; Vars: <i>code_module, code_presentation, id_assessment, assessment_type, date, weight</i>	Understanding course design - assessments structure and weight. *Links to studentAssessment by id_assessment
studentVle	Core LMS usage log: daily count of clicks per student; Largest table (millions of rows); Many students have 0 or few clicks; Skewed distribution Dim: $10,655,280 \times 5$; 32.8MB; Type: Time-series click data; Vars: <i>code_module, code_presentation, id_student, id_site, date, sum_click</i>	Engagement analysis, building learning behavior profiles, linking LMS activity to outcomes; * Very large; many students have 0 or few clicks; skewed distribution

Vle	Metadata on VLE materials (resource type: forum, quiz, wiki, reading, etc.) Dim: 22 × 3; 264.3KB; Type: Course metadata; Vars: <code>code_module</code> , <code>code_presentation</code> , <code>module_presentation_length</code>	Contextualizing studentVle activity logs: kind of resources, clicks reference * Small reference table
Unusual features: Dates often negative relative to course start; ‘?’ used in place of missing date values; aggregation needed for <i>studentVle</i> , <i>studentAssessment</i> to avoid duplication, and there are highly imbalanced outcomes due to withdrawals.		

2.2. Dataset Origin & Source: While other reputable repositories (*xxx.gov*, *Kaggle*) could have been used, the decision to use the University of California’s (UCI) OULAD was because it is publicly available and well structured. Its data was birthed from a 2013-14 academic year survey from a UK open university, consisting of 22 course presentations, about 32,593 students, and 10 million records of VLE interactions [OpenAI, 2025].

2.3. Original Collection Purpose: Though released in 2015, years after its capture, the public dataset was purposed to support research on student performance, LMS retention, and LMS interaction patterns.

2.4. Key Variables and Their Importance:

Variable	Importance
StudentInfo(<i>gender</i> , <i>region</i> , <i>age_band</i> , <i>highest_education</i> , <i>imd_band</i> , <i>disability</i>)	Capture demographics & equity factors; Strong predictors of engagement, withdrawal, and achievement;
<i>Final_result</i>	Target variable for most predictive models (Withdrawn / Fail / Pass / Distinction)
StudentVle+vle(<i>sum_click</i> , <i>activity_type</i> , <i>week_from</i> , <i>week_to</i>)	Provides direct evidence of learning outcome; Timeliness of submissions signals persistence; Weighted scores approximate overall grade
studentAssessment+Assessment(<i>score</i> , <i>date_submitted</i> , <i>assessment_type</i> , <i>weight</i>)	Gives highest level prior to registering (<i>No formal qualifications</i> , <i>Lower Than A Level</i> , etc.)

3. DATASET SOURCE & RELEVANCE

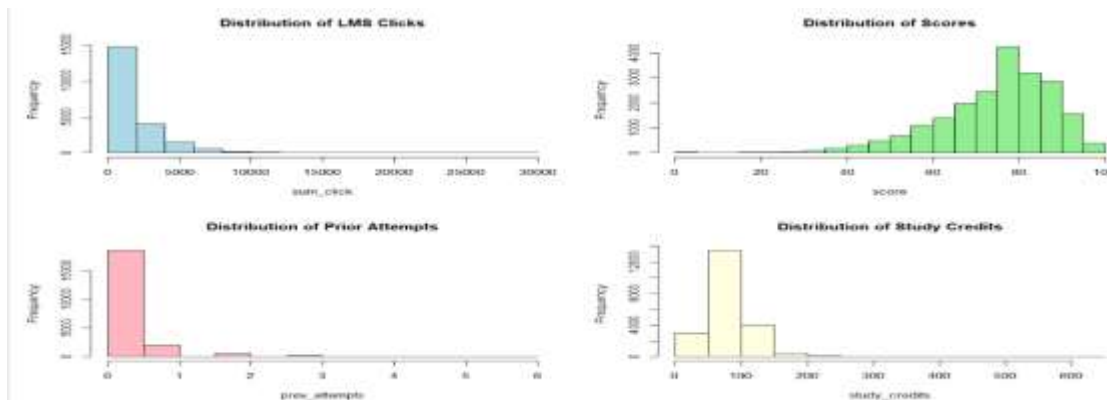
3.1 Dataset Source’s Reputation: This is famed because it is widely cited and used. It is clean, structured, analyses-ready, and perfectly aligned for any research for both student performance and LMS usage. It is further respected for both learning analyses and academic researches

3.2 Dataset Relevance to Research Question: OULAD is relevant because it can be used both for LMS activity logs and student performances. Given the data capacity, it can provide higher statistical power than other datasets (*EdNet*, *OpenDataBay*) [OpenAI, 2015]. Further, the flexibility afforded with the selection of any combinations of tables can be managed based on the research interest. Only the **Course** table (info. on course modules) will be omitted. While surface analyses could be obtained from the *studentInfo* and *studentRegistration* tables, the combination of *studentAssessment*+ *assessment*, and the *studentVle*+*vle* tables would generate more granular insights for extended research. Also, the identification of linking fields to merge datasets can be easily achieved. [Jawthari, M. et al, 2022] held that behavioral variables (*engagement*+*assessments*), gave the greatest predicative models; demographic variables added interpretive context, and registration variables refined risks opportunities.

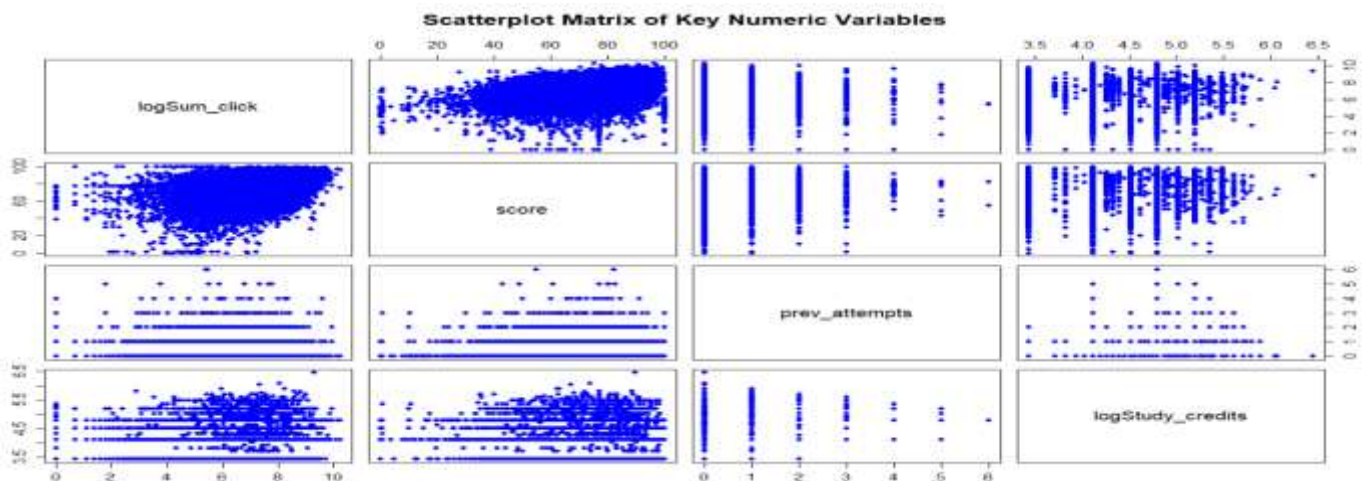
4. EXPLORATORY DATA ANALYSIS

Interpretation of Plots: While specific analyses are captured in the markdown file, only overall summary will be given for specific plots. Corresponding figures immediately follow relevant sections.

4.1 Histograms (Fig. 1): These were chosen to show the spread, skewness, and outliers of the numeric variables. Overall, engagement and workload vary widely; most students are new enrollees; performance is broadly distributed but not extreme. Specific plots: **Sum_click**: Right-skewed → most students have low-to-moderate LMS activity; few with very high engagement. **Score**: Roughly symmetric → grades spread fairly evenly, suggesting no strong skew in performance. **Prev_attempts**: highly concentrated on [0, 1] → most students are first-timers, and **Study_credits**: Slightly right-skew → most students enroll in moderate credit loads, fewer in very high ones. Figure 1 follows.

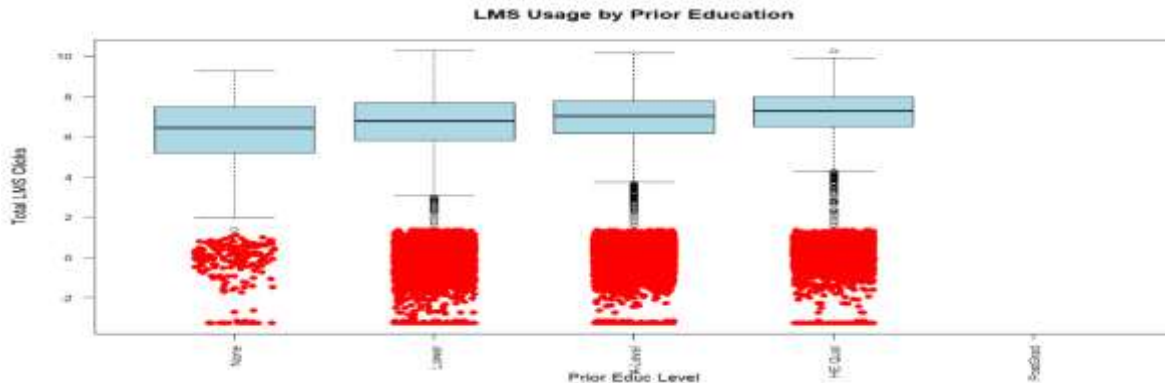


4.2 Scatterplot Matrix (Fig. 2): This plot is used to show the relationship of numeric variables to each other, potential moderators, and interaction patterns for modeling. Summarily, the visualization supports that **student engagement** is the strongest behavioral correlation of achievement (*score*), while *prev_attempts* acts as a potential risk indicator. A clear positive, nonlinear trend is visible → higher LMS activity corresponds to higher scores, though the effect plateaus for higher scores, supporting the notion that engagement predicts performance, but with diminishing returns. An inter-correlation view between *logSum_click* and *logStudy_credits* shows: mild positive association → students with more credits may interact more frequently with the LMS, perhaps reflecting greater academic engagement overall. Figure 2 follows.

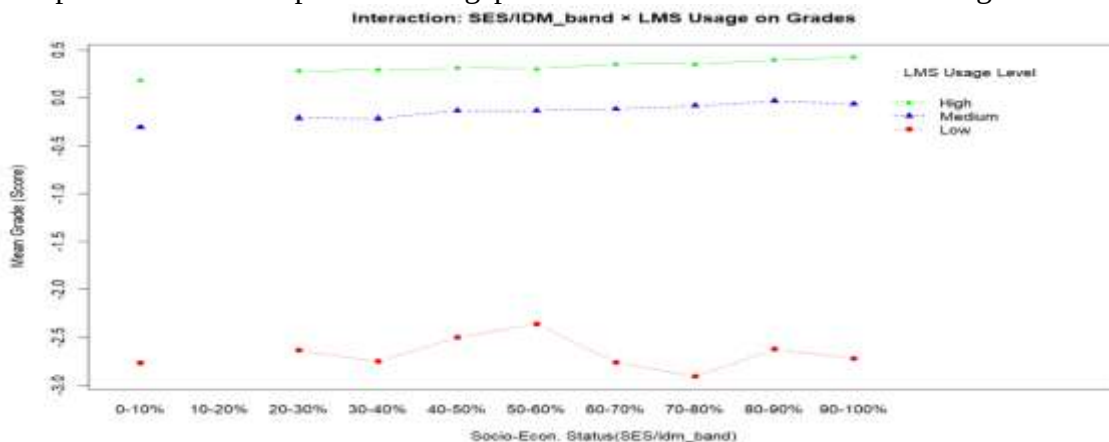


4.3 Boxplot (Fig.3): This plot investigates whether prior education, SES, or age affects engagement and shows the log-LMS activity across five levels of prior education. Log transformation successfully mitigates extreme right-skewness, allowing clearer comparison among groups. Overall, the *median LMS engagement*

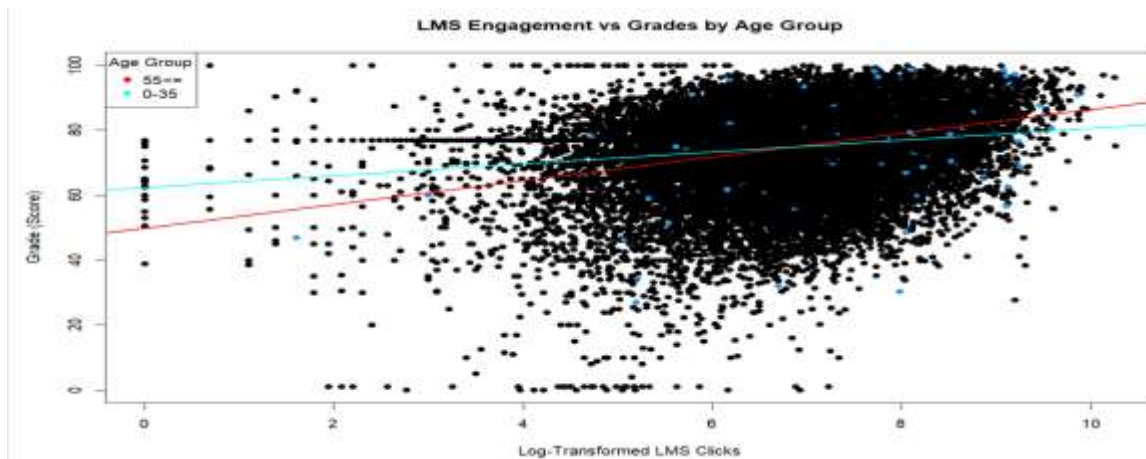
increases modestly with education level → students with higher education show slightly higher usage, though the variance within each group is substantial. The compressed lower whiskers and numerous outliers below zero suggest that *non-engaged or minimally active learners are common across all education levels*, particularly among those with lesser qualifications. The overlapping interquartile ranges indicate *no sharp distinction in engagement by prior education* → prior academic attainment may not strongly predict LMS activity. However, the trend toward greater median activity among more educated groups suggests *more consistent study habits or digital literacy advantages* at higher levels. Figure 3 follows.



4.4 Interaction Line Plot (Fig. 4): The question ‘are students from lower SES disadvantaged even with high LMS engagement is resolved. The overarching insight: The interaction demonstrates that LMS engagement exerts a dominant effect on academic success, stronger than even SES. High engagement appears to buffer against socioeconomic disadvantage, supporting the hypothesis that digital engagement mediates educational inequality. Summarily: 1) **Main effect:** Higher LMS usage → higher grades; 2) **Moderate effect:** SES moderates this effect weakly; engagement benefits all SES levels but mostly improves outcomes for lower SES students, and 3) **Pedagogical implication:** Encouraging consistent LMS participation can narrow performance gaps linked to socioeconomic status. Figure 4 follows.



4.5 Scatter Plot+Regression Line (Fig. 5): To examine how LMS engagement affects grades and how this relationship varies by demographics, the **scatter plot with regression lines** is most appropriate. It illustrates how grades change continuously with LMS clicks and shows moderation effects—different slopes across age groups reveal that the impact of engagement varies by age. Transforming the *click* variable, corrects right-skewness, stabilizes variance, and improves interpretability. Overall, **positive slopes** indicate that greater LMS interaction generally aligns with higher grades, though variations in slope and data spread suggest age-related differences in engagement benefits across ages and learning behaviors [Papamitsiou, Z. et al, 2014]. Caution is warranted: correlation does not imply causation—high engagement may characterize both motivated and struggling learners. Figure 5 follows.



4.6 Missingness: Most missing data (2,839) occur for the *stud_unreg* variable – possibly so because of withdrawals. Since the analysis is not focusing on withdrawn engagements, these rows have been removed. Variables with moderate amount of missing values age (possibly because females consider this confidential), and SES (identifying with this or education status could be both stigmatizing and embarrassing for underachievers). These missing values are resolved by using their respective modes. Missing scores (173), replaced by their medians, since its distribution is skewed.

4.7 Outliers and Distributions: From Fig. 3, the outliers indicate students who are unusually high or low in engagement relative to their group. Since our study emphasizes behavioral patterns and engagement, weighting or covariate adjustment is preferable over discarding data. Hence, outliers are presumed to represent *legitimate high-engagement students* rather than data errors. In respect to the shapes of distributions, removing outliers greatly enhanced analytic interpretation. Another concern, imbalance (gender), will be resolved by resampling (oversampling/undersampling) which may be more relevant for Inferential Analyses. As a result, two datasets will facilitate this project: one for descriptive reporting and another for inferential insights (synthetic). The *sum_click* and *study_credits* distributions were right skewed but resolved by log transformation to yield clearer visual trends.

4.8 Missing Data Assessment: The *date_reg* variable contains some NAs while the *data_unreg* variable contains '?' as missing values. For categorical variables, missing data are filled by mode values; otherwise, since the score variable is negatively skewed, the median is substituted.

4.9 Outlier Detection: Outliers (from fig. 3) are addressed by log transformation and are assumed to be valid behavioral variations, not measurement errors. Through log transformation and standardization, their influence is managed statistically rather than eliminated thus preserving analytical integrity and representativeness across the full engagement spectrum.

4.10 Distribution Analysis: The LMS usage variable is strongly right skewed, typical of engagement data with few highly active users. Applying log transformation normalizes the spread, reduces skewness, and stabilizes variance, improving suitability for comparison and analysis. *Scal_score* displays an approximately normal pattern centered near 0. Overall, the transformation is effective, preserving behavioral variation while meeting assumptions of linearity and homoscedasticity.

5 METHODOLOGY

Addressing the main question, *can demographic and LMS variables explain achievement gaps?* the methodology is outlined below.

5.1 Overview and Purpose: The modeling technique employed in this study is a **Multiple Linear Regression (MLR) model**, designed to examine the relationship between several categorical and numerical predictor and a continuous outcome variable — the **scaled score (*scal_score*)**. This approach is primarily used for **prediction and inference**, allowing the estimation of how various individual, regional, and socioeconomic factors contribute to differences in the educational performance.

MLR extends the **simple linear** model by incorporating multiple independent variables, each representing a distinct influence on the outcome. It estimates how each predictor affects the response while controlling for all others - providing **marginal effects** for both categorical and continuous inputs. This makes it a suitable technique for understanding patterns of performance or behavior across diverse populations.

The present model initially integrates both **categorical predictors** — *regional grouping (*reduceReg*)*, *code_mode*, *gender*, *age_band*, *disability*, *highest education (*high_educ*)*, *SES (*imd_band*)*, and *final results (*fin_res*)*, the only **numerical predictor**, *logSum_click*, representing user engagement. Together, these variables capture structural, demographic, and behavioral dimensions that influence performance outcomes.

MLR is especially useful in this context because:

1. It enables **simultaneous assessment** of multiple explanatory factors.
2. It provides **quantitative estimates** of how changes in predictors correspond to changes in the outcome.
3. It supports **hypothesis testing** on whether specific predictors have statistically significant effects.
4. It allows for **adjusted interpretations**, meaning each coefficient reflects a unique contribution of other variables. These usefulness were cited by [Al-Bahrani et al, 2020].

The general form of the MLR equation: $v(\rho, x_{\{1\}}, x_{\{2\}}) = \beta_{\{0\}} + \beta_{\{1\}x_{\{1\}}} + \beta_{\{2\}x_{\{2\}}} \dots + \beta_{\{n\}x_{\{n\}}}$ (1) where

v is the independent variable (*scal_score*), $x_1, x_2, \dots, x_n$ are the independent variables, β_0 is the intercept (constant), and $\beta_1, \beta_2, \dots, \beta_n$ are the regression coefficients for each independent variable. Additionally, the coefficients for the predictors can be found by matrix algebra:

$$B = (XX)^{-1}XY \quad (2) \quad \text{as supported by [K. Nailaizza., et al., 2024, Roustaei N., 2024].}$$

Specifically, our regression model can be expressed as:

$$mlr_model \leftarrow lm(scal_score \sim reduceReg + gender + age_band + disability + high_edu + imd_band + finRes + logSum_click + code_mode, data = df) \quad (3)$$

or mathematically as:

$$scal_score_i = \beta_0 + \sum_{k=2}^K (\beta_k D_{ik}^{reduceReg}) + \beta_g D_i^{gender} + \beta_a D_i^{age_band} + \beta_d D_i^{disability} + \beta_h D_i^{high_edu} + \beta_m D_i^{imd_band} + \beta_f D_i^{fin_res} + \beta_c D_i^{code_mode} + \beta_x logSum_click + \varepsilon_i \quad (4)$$

where:

- **scal_score_i** = outcome for individual i
- **β_0** = intercept
- **β terms** = regression coefficients for predictors
- **D terms** = dummy-coded categorical predictors
- **logSum_click** = numeric predictor

- ε_i = random error term

This formulation allows the model to capture both the **mean differences** between categorical groups and the **linear effects** of the continuous variable, offering a robust framework for both **predictive analytics** and **statistical inference**.

5.2 Type of Prediction: Goal is to **predict continuous student performances (scal_score) based on demographics, and LMS activity**.

A simpler representation of MLR: $Y_i = \beta_0 + \sum_{j=1}^p \beta_j X_{ij} + \varepsilon_i$ where Y_i is the outcome score for student i , X_{ij} = j -th predictor for student i , β_j = coefficient for predictor X_j and $\varepsilon_i \sim N(0, \sigma^2)$ = independent error term normally distributed with mean 0 and variance σ^2 .

5.3 Model Components and Operation: The modified MLR model operates by estimating how a set of predictor variables collectively influence a continuous variable—*score*. Each predictor is assigned a coefficient (β) that quantifies its average effect on the outcome when all other variables are held constant.

Categorical variables: reduceReg (20+ levels), **gender (M/F)**, **age_band** (Ordered rank: 0-35, etc.), **disabil (Y/N)**, **high_educ (Y/N)**, **imd_band (socio-economic status)**, **code_mode (AAA, BBB,...)**, **finRes (pass, fail)** and **logSum_click** (log sum of clicks - numerical).

The algorithm estimates coefficients by minimizing the **sum of squared residuals (least squares)**: $\min \sum_i (\text{scal_score}_i - \hat{y}_i)^2$, where $\hat{y}_i = \mathbf{X}_i \boldsymbol{\beta}$ is the predicted value. This process identifies the hyperplane that best fits the data in multidimensional space. The magnitude and sign of each β indicate the strength and direction of the variable's influence on performance outcomes. This modeling approach not only captures **linear associations** but also enables **interpretability**, for clear inference about which learner attributes most strongly predict success.

5.4 Rationale for Algorithm Selection: MLR is chosen because the primary outcome, *score (scal_score)*, is continuous, and the research goal is to **quantify the contribution of each predictor to performance**. Unlike classification techniques (**Logistic Regression, Decision Trees**), MLR directly estimates how changes in predictors relate to changes in a numeric outcome, producing **interpretable coefficients, β** , with clear meaning. Compared to **Random Forest Regression** or **Gradient Boosting** which also predicts continuous values, MLR offers: 1) **Transparency and interpretability** – coefficients indicate effect size and direction, enabling actionable insights; 2) **Simplicity and computational efficiency** – easy to implement and validate on moderate-sized datasets; and 3) **Statistical inference** – allows hypothesis testing (T-tests for coefficients, ANOVA), and Confidence Intervals for effect estimates.

Despite the constraints listed below, MLR is well-suited for studies aiming to **understand relationships rather than only predict outcomes**, making it the preferred approach.

5.5 Assumptions

- Linearity:** Relationship between numerical predictors and outcome is linear.
- Independence:** Observations are independent.
- Homoscedasticity:** Residuals have constant variance.
- Normality:** Residuals are approximately normal (for inference).
- Non-perfect multi-collinearity** among predictors.

5.6 Constraints and Limitations

- Categorical predictors:** Factors with many levels may inflate variance.
- Category sample size:** Each level must have enough observations to reliably estimate effects.
- Missing data:** Observations with NAs would be handled via removal or imputation.
- Linearity:** Relationships between predictors and outcome must be approximately linear.
- Independence of observations:** Non-independent data can bias estimates.
- Homoscedasticity:** Unequal residual variance may reduce efficiency and validity of p-values.
- Multi-collinearity:** Highly correlated predictors inflate standard errors, affecting interpretability.
- Outliers/influential points:** Extreme values can distort coefficient estimates.

5.7 **Awareness of Limitations Affects:** This is essential for responsible interpretation of the MLR model. Given the assumptions of the MLR, violations can bias estimates or weaken predictive accuracy. Recognizing these limits helps prevent overconfidence in results and guides appropriate data transformation or model revision. Ultimately, awareness fosters transparency and strengthens the credibility of analytical conclusions.

5.8 Benefits

- Standard, interpretable model for continuous outcomes.
- Coefficients indicate effect of each predictor (relative to reference levels).
- Can **easily handle both categorical and numerical predictors**.
- Basis for adjusted predictions, variable importance, and policy insights.
- Provides **quantitative estimates of effect size** for each predictor.
- Accommodates **mixed variable types** (categorical and numeric).
- Useful for **policy or educational interventions** - identifying underperforming regions or demographic groups.
- Scaled outcomes allow comparability across cohorts.

5.9 **Critical Insights:** Based on [Romero, C. et al, 2020], evaluation emphasizes the balance between interpretability and predictive accuracy in MLR models. Simpler models improve transparency and communication but may miss complex nonlinear patterns, while more intricate models capture nuance at the expense of clarity. Effective model choice depends on research goals, not just statistical performance. Practically, MLR's transparency enables actionable insights and policy relevance, though its assumptions must be recognized to prevent over-generalization. Ultimately, achieving the right balance between complexity and interpretability ensures both rigor and real-world usefulness.

6. MODEL BUILT

6.1 Model Construction & Evaluation

Alternatively, the MLR predictive model can be specified as: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$ (5) as supported by [Nailaizza, K. et al, 2024]. This MLR model was upgraded to the GBM then eventually to the LightGBM of the form: $\hat{Y} = \sum_{m=1}^M \lambda_{\{T_m\}(X)}$.

The latter has main advantages of handling **larger datasets, more categorical features, and faster grid search** due to optimized histogram-based algorithms.

- a. **Data Partitioning & Cross-Validation (CV):** Instead of a single train/test split, the dataset is partitioned into 5 equal subsets. Each fold takes turn serving as a validation set, while the remaining 4 folds form the training set. This ensures every observation contributes to both model training and evaluation. For each fold $f \in \{1, 2, \dots, 5\}$: $Lf(\beta) = \sum_{i \in \text{train}f} (y_i - \hat{y}_i)^2$ (6)
- b. **Model Estimation per Fold:** For each training subset, parameters are estimated by minimizing the **sum of squared residuals**: $\beta = (X^T X)^{-1} X^T y$. This yields optimal coefficients $\beta = \{\beta_0, \beta_1, \dots, \beta_p\}$ for that fold [Nailaizza, K. et al, 2024]. Predicted outcomes for the validation fold are: $\hat{y}_i = x_i^T \beta$ (7)
- c. **Model Performance per Fold:** This is evaluated using **Root Mean Squared Error (RMSE)** for each fold f : $RMSEf = \sqrt{((1 / nf) \sum_{i \in \text{val}f} (y_i - \hat{y}_i)^2)}$ (8); nf is the number of valid observations in fold f .
- d. **Aggregated Model Performance:** After all folds are completed, average performance is computed as: $RMSE_{cv} = (1 / k) \sum_{f=1}^k RMSEf$. This represents the model's expected predictive accuracy on unseen data and provides a stable, low-variance estimate of generalization performance.
- e. **Coefficient estimates (β)** represent the expected change in the dependent variable for a one-unit change in each predictor, with all other predictors constant.
- f. **Cross-validated RMSE_{cv}** quantifies the average prediction error.
- g. **Standard deviation of RMSE_f** across folds measures stability.

6.2 Key Parameters

- a. **MLR: Regression Coefficients $\beta_{\{j\}}$** (represent the contribution of each predictor during via OLS), **Intercept β_0** (captures the expected outcome when all predictors are 0 providing a baseline reference), and **Error Variance σ^2** : (measures variability of residuals and influences CIs and prediction intervals) [Roustaei, N. et al., 2024].
- b. **LightGradient Boosting Model(LBM/GBM):** **ntrees** (number of iterations -more trees = more learning, but higher risk of overfitting, **interaction.depth** (maximum depth of each tree), **shrinkage** (learning rate controlling how much each tree contributes-smaller values = slower, more stable learning, and **n.minobsinnode** (minimum number of observations required in a terminal node).

6.3 Parameter: Interpretation and Influence

- a. **Magnitude and Sign of $\beta_{\{j\}}$** : Directly indicate effect direction and strength [Roustaei, N. et al., 2024].
- b. Parameter interpretation for GBM is presented above in the **Key Parameters** section.
- c. **Tuning and Validation Guidelines**
 - Use **5-fold CV** to assess predictive performance.
 - Evaluate performance metrics (RMSE, AIC, R^2 , Adj. R^2 , MAE, Avg. runtime) on held-out data to avoid overfitting. AIC and Adj- R^2 are not applicable to the LBM/GBM.
 - Parameter estimates are refined iteratively using training data while ensuring unbiased generalization to test data. Grid Search varies parameters to find the best parameters.
 - Variables from the full model whose p-values are significant are piped into a reduced model. Then the Winsorized approach where 5% of the top and bottom observations are trimmed to reduce the influence of extreme values by capping them at specified percentile limits. This aligns the data to its original shape preventing outliers from disproportionately skewing estimates and improving stability and robustness.
 - Variables: *code_mode*, *high_educ*, *imd_band*, *Sum_click*, and *finRes*, **initially have** significant influence on student scores.

6.4 Modeling, Training & MLR Connection

The linear model learns the best-fitting weights that describe how predictors (e.g., *code_mode*, *gender*, *disabil*) contribute to **scal_score**, estimating coefficients through least squares and minimizing prediction error. Cross-validation evaluates how stable and generalizable these coefficients are across different subsets of the data.

During training, the **MLR model**:

- Estimates coefficients β_j describing each predictor's linear contribution of **scal_score**.
- Minimizes SSE to reduce prediction error.
- Adjusts for correlations among predictors to prevent over-attribution.
- Provides uncertainty metrics such as standard errors, t-values, and confidence intervals.

To improve predictive performance beyond linearity, **Gradient Boosting Model (GBM)** with 5-fold CV and grid-search tuning iteratively builds an ensemble of trees, where each tree corrects the residual errors of the previous one(s). Tuning parameters such as the *learning rate*, *number of boosting iterations*, *interaction depth*, and *minimum node size* help balance accuracy and overfitting. The GBM's model is of the form: $\hat{Y} = \sum_{m=1}^M \lambda_{\{T_m\}(X)}$.

A **winsorized interaction model** further strengthens performance by reducing the influence of extreme outliers while capturing combined effects between variables, improving stability and predictive accuracy compared to a simple linear model. Thereafter, the **LightGBM** model, a variant of GBM using the same variables as the best GBM (model6) is explored, with resulting parallel performance.

5-Fold CV repeats this training across five partitions, producing fold-specific coefficients, predictions, and validation errors. The aggregated results estimate how well the model generalizes to unseen data.

6.5 Validation Strategy and Model Generalization

To assess how well the model generalizes with unseen data using **5-fold CV**, consideration was made for:

Appropriateness:

- Efficient use of data:** Every observation is used for both training and validation, which is important when the dependent predictor is transformed and the dataset has missing values.
- Reduces overfitting bias:** By validating on unseen folds, the model's predictive ability is tested on unfamiliar data.
- Stable estimates:** Averaging across folds mitigates the variability that can occur if a single test set is used.

Context relevance: The model derives scores from multiple predictors. CV ensures that performance metrics reflect real-world performance across different learning modes, gender, and demographic groups. Thus, the **5-fold CV** provides a **robust, unbiased estimate of model performance** and generalization for datasets with mixed variable types, transformed predictor, and potential missingness. It is particularly suitable because **it tests the model across multiple subsets of the data** rather than relying on a single split, giving confidence that the learnt relationships are stable and meaningful.

6.6 Performance Metrics and Interpretation

To evaluate the predictive quality of the model, the following metrics are calculated:

- Root Mean Squared Error (RMSE):** $RMSE = \sqrt{(1/n) \sum_{i=1}^n (y_i - \hat{y}_i)^2}$. It measures the average magnitude of prediction errors in the same units as *scal_score*. Lower RMSE indicates better

predictive accuracy. In CV, the Mean RMSE across folds reveals the model's typical prediction error.

- b. **Mean Absolute Error (MAE):** $MAE = (1/n) \sum_{i=1}^n |y_i - \hat{y}_i|$ - is the average absolute deviation of predictions from observed values, providing a robust measure less sensitive to outliers than RMSE.
- c. **Coefficient of Determination:** $R^2 = 1 - (\sum_{i=1}^n (y_i - \hat{y}_i)^2) / (\sum_{i=1}^n (y_i - \bar{y})^2)$. R^2 quantifies the proportion of variance in *scal_score* explained by the model. Values close to 1 indicate high explanatory power. For CV, the mean and median R^2 across folds indicate generalization performance.
- d. **Akaike Information Criterion (AIC):** $AIC = 2k - 2 \ln(L)$: $\hat{L}$ is the maximum likelihood of the model. It balances model fit and complexity. Lower AIC indicates a better trade-off between goodness-of-fit and simplicity.
- e. **Average Runtime (s):** Averages computational time to fit the model per fold in seconds. While not a measure of accuracy, s, provides insight into computational efficiency, for larger datasets [Uyanik, G.K. et al., 2013].

7. PROJECT RESULTS

7.1 Model Building in R

The entire workflow is implemented mainly in base R, fully reproducible with a fixed seed (123) and 5-fold cross-validation. After data cleaning, all predictors are included initially, then iteratively reduced by removing insignificant variables. Detected outliers are handled through Winsorization (5% –95%). Because MLR cannot handle single-level factors, the data is subsetting accordingly. Despite trimming, many predictors remain insignificant, prompting exploration of grouped variables, interactions, and additive terms. This modeling strategy was supported by [Tsiliki, G, et al, 2015]. From the notebook, assumption checks are conducted prior to Model 3.

Due to limited usefulness of MLR, a Gradient Boosting Model (GBM) is adopted using 5-fold CV, and a Grid Search approach for exploring the best parameters. Thereafter, to further exploit this potential, its counterpart, the LightGBM is employed. Predictors with low relative influence are not automatically discarded. However, non-informative variables are removed progressively. Model 6 uses moderate tuning and ultimately outperforms other subsequent models. From the R notebook, Model 8 shows that *finRes* and *logSum_click* are the strongest drivers of scores; however, using only these factors worsened predictive accuracy. [Friedman, J. H, 2001] commented on GB's model with CV. The equivalent LightGBM model, using the same building blocks lived up to expectation.

Model robustness is tested with 1,500 synthetic data. **Winsorized data under cross-folding further improves fit.** Table 1 tracks model development and results and shows progress of model variation for the reported results. *Columns 6 and 10* are reported; column 5 is not included because no additional insights are seen. Missing values for *col. 10* (AIUC, Adj-R²) are meaningless and not applicable.

Table 1

Metrics	Original Full MLR (Model1)	Original Reduced MLR (Model2)	Winsorized Sub-setting MLR (Model3)	Winsorized Interaction MLR (Model4)	GBM: Winsorized No Interaction – (gend, disabil) (Model6)	GBM: Winsorized, Reduced Par, normal parameters (Model7)	GBM: Winsorized, Less Params, Increased Param tune (Model8)	GBM: Winsorized Increased Grid Params, Iteration (Model9)	LightGBM: Winsorized, 5CV+ Grid Search (Model10)
Residual Std. Error	0.7317	0.7317	0.444	0.7069					
Multiple R ²	0.4808	0.4805	0.1485	0.4956	0.6161	0.5659	0.558		0.6060
Adjusted R ²	0.4799	0.4799	0.1004	0.4949	0.5017				NA

Metrics	Original Full MLR (Model1)	Original Reduced MLR (Model2)	Winsorized Sub-setting MLR (Model3)	Winsorized Interaction MLR (Model4)	GBM: Winsorized No Interaction – (gend, disabil) (Model6)	GBM: Winsorized, Reduced Par, normal parameters (Model7)	GBM: Winsorized, Less Params, Increased Param tune (Model8)	GBM: Winsorized Increased Grid Params, Iteration (Model9)	LightGBM: Winsorized, 5CV+ Grid Search (Model10)
Mean RMSE	1.2323	0.7324	0.4831	0.7077	0.7090	0.6467	NA	0.6558	0.6164
SD RMSE	0.0238	0.0098	0.0552	0.0127	0.0211	0.0142	NA	0.0108	0.0119
Mean R ²	-0.5177	47.81	0.0183	49.35	60.60	55.33	55.8	55.40	60.61
Median R ²	-0.5110	0.477	0.0601	0.4938					
Mean AIC	32251.32	32241.54	414.64	31236.07	31342.57				NA
Avg Runtime	0.0295	0.0197	0.0032	0.0277	155.4453	116.1819	64.6391	145.4579	84.09
Mean MAE	0.8935	0.5379	0.3967	0.5215	0.4304	0.4679	0.4625	0.4627	0.4305
SD MAE	0.0159	0.006	0.0474	0.0069	0.0087	0.0093	0.0048	0.0065	0.0084

Interpretation (Model 10):

1. **Accuracy improved** with winsorized reduced model: lower RMSE and MAE, and higher R².
2. **Stability improved:** SD of RMSE and MAE decrease, showing more consistent predictions across folds.

Winsorizing the response variable and using a focused reduced model improve predictive power, fit, and stability compared to both the original full and reduced models. The winsorized GBM models **outperform the winsorized MLR model (with interaction)** in predictive accuracy (lower RMSE/MAE, higher R²). Both GBM capture nonlinearities and interactions more flexibly than the MLR.

7.2 Plots and Visualizations (R Notebook)

MLR assumptions are checked using standard diagnostic plots (*Residuals vs Fitted*, *QQ*, *Scale–Location*) and all are generally satisfied. A fold-wise RMSE plot shows across folds, indicating uneven stability. Boxplots reveal many outliers—that are later winsorized—and shows that code mode and academic status create the clearest score differences, while gender effects remain small. Sub-setting disability and pass groups show shifts in score variability within specific populations.

Engagement effects are especially strong: the *logSum_click* boxplot quartiles display a clear monotonic rise in assessment scores with higher platform interaction. Marginal effect plots confirm that predicted scores increase with clicks, with diminishing returns at high activity levels and narrow confidence bands across most of the range. Overall, visual evidence across the figures shows that click engagements are a strong, nonlinear, and reliable predictor of performance.

7.3 Accuracy

The following accuracy metrics are reported:

Residual Std. Error	Measures the typical size of prediction errors in original units. Lower is better — indicates tighter fit to the data.
Multiple R ²	Tells how much variance in the dependent variable, <i>scal_score</i> , is explained by the predictors, highlighting explanatory power. Higher R ² (closer to 1) indicates better explanatory power. Fold-wise averaging accounts for potential variability across CV splits.
Adjusted R ²	Like R ² , but penalizes adding weak predictors. Better for comparing models with different numbers of variables (MLR).
Mean RMSE	Average prediction error across folds. Shows how well the model generalizes , not just fits training data.
SD RMSE	Variability in RMSE across folds. Low SD → model performs consistently , not just on certain splits.

Mean R ²	Average explained variance across folds. Confirms model strength beyond the original sample.
Median R ²	Less sensitive to extreme folds. Shows “typical” performance across data splits.
Mean AIC	Allows model comparison, balancing fit and complexity — useful if considering alternative models or regularized approaches. Lower AIC indicates a better trade-off between fit and simplicity. Useful for comparing alternative models in future work (MLR).
Avg Runtime	Shows computational efficiency. Low runtime → easier to re-train, deploy, or iterate.
Mean MAE	This is easier to interpret than RMSE. Represents typical prediction missed in dependent variable.
SD MAE	Measures how stable the absolute error is across folds. Low SD → model is predictably accurate .
Taken together, these metrics show not just how well the model fits the data (using R ² , RMSE) but how well it generalizes (CV RMSE/R ²), how it stabilizes (SD metrics), how it is efficient (runtime), and whether it is parsimonious (AIC)	

7.4 Model Generalization

To assess the generalization of the model, 1500 random synthetic data values are generated. Cross-validation is used, so that performance could be tested on multiple held-out folds rather than training data alone. Further, the GBMs handle *non-linear patterns and interactions* automatically, reducing under-fitting according to [Kumar, S., et al, 2025]. Finally, the model consistently selects the same variables (*logSum_click*, *finRes*, *gender*×*disabil*), indicating stability across folds.

7.5 Model Robustness

Cross-validation consistency

RMSE, MAE, and R² remain stable across folds (low SD), showing that model performance does not depend heavily on which subset of training data is used.

Random seed sensitivity

While MLR can shift slightly with different seeds, the tuning loop repeatedly converges on similar hyper-parameters, indicating consistent behavior.

Outlier sensitivity

GBM handles outliers better than the MLR but still reacts to extreme values—especially in predictors like *logSum_click*. Thus unusual engagement patterns can influence predictions.

Categorical imbalance sensitivity

Rare category combinations (*gender* × *disability* × *finRes*) may cause local instability, though global patterns remain stable.

Overall stability

The model is reasonably stable with only moderate sensitivity to outliers and rare categories.

Reliability

Cross-validation metrics show low variability, meaning predictions are generally trustworthy and not driven by peculiarities in single folds.

Remaining Risk: The complex interaction term (*gender* × *disabil* × *finRes*) and multiple categorical levels increase the chance of **localized overfitting** where some rare subgroups may be poorly learned.

Overall: The model appears **moderately reliable** with **controlled overfitting**, but rare categories and noise remain the main vulnerabilities. **A reduced model is preferable—retaining the key predictive effects while simplifying interpretation, without sacrificing accuracy or overall fit.**

8. CONCLUSION

8.1 Summary and Interpretation of Results: The MLR model is firstly employed to predict *scal_score* using categorical and continuous predictors. Various selection of predictors is used to build the model but those not contributing to its strength were dropped. The best MLR model achieved: $R^2=49.35$ explainability in 0.028 secs. **Eventually, the combination of GBM, 5-fold CV, Winsorization and variable mixings (additive, interactive) are used to boost predictive power.** The GBM's results are accepted (col. 10) considering **the dataset involves patterns of human behavior** resulting in *logSum_click* (68%) and *finRes* (21%) being the greatest contributors to student scores. When the computations results are compared to other MLR models, the LightGBM model performs better and is comparable to its counterpart, the GBM. The former emphasizes behavioral and structural factors over demographic characteristics for predicting student achievement with the LightGBM providing clearer prioritization of impactful predictors through its gain-based importance metric. **This insight can guide targeted interventions:** boosting engagement (*logSum_click*) or program-specific support (*code_mode*) may yield the greatest improvement in performance outcomes.

LightGBM results: Mean RMSE= 0.6164, $R^2=60.60$ in ~84.09 seconds, indicating that predicted values are on average, very close to observed scores. Fold-to-fold variability (SD RMSE = 0.0119) reflects moderate sensitivity to specific data partitions, particularly those with rare factor levels. Residual diagnostics largely confirm linearity, homoscedasticity, and approximate normality, supporting the validity of model assumptions. Categorical predictors, *code_mode* and *imd_band*, show clear differences in median *scal_score*, visualized via boxplots, while continuous predictors contribute linearly.

Datasets: Two datasets are produced—one cleaned and one strictly for prediction. The latter would yield better performance due to larger size, variability, and more representative coverage of predictors, enabling more stable coefficients and higher explanatory power.

8.2 Interpretability:

The model demonstrates strong predictive structure and behaves consistently across folds, **but the synthetic data predictions should be interpreted cautiously.** It produces a **centered distribution around zero**, with a **mean=-0.7424** and a **median=-0.4335**. The negative skew suggests that the model predicts slightly lower than the scaled average score for most cases. The range (-4.4694, 1.3245) indicates that some extreme synthetic combinations lead to lower predicted scores, while a few reach modestly high values. Because the synthetic 1,500 observations are not real, they cannot validate accuracy or real-world performance; instead, they only reveal how the model behaves mathematically. [Shmuelio, G., 2010] cited that these predictions are still useful for diagnosing stability, checking reasonable ranges, and examining directional effects, but they do not constitute evidence of actual student outcomes.

8.3 Practicality:

The GBM models (6 & 10) are practical and effective for predicting *scal_score*, offering stronger flexibility and predictive accuracy than linear models while remaining computationally manageable for larger datasets. [Kabakchieva, D., 2013] supported the model's ability to capture interactions and non-linear effects suited for educational data, where interpretability is supported through variable importance and partial-dependence diagnostics. Synthetic data testing shows stable model behavior. Though void of real-world validity, ethical use requires care with sensitive predictors. Overall, *model10* provides a feasible, robust, and well-structured approach to modeling student performance. The winning model: **model10 $\leftarrow$ gbm(formula = Wscal_score ~ code_mode + gender*disabil*finRes + high_educ + imd_band + finRes+logSum_click).** **proj_predData.csv** is a ready-to-use dataset on which predictions can be made.

8.4 Identification of Weaknesses

Despite moderate predictive performance, several performance limitations for MLR were highlighted by [Baker, R. et al, 2014]:

1. **Data Quality and Completeness:** Missing values and rare categorical levels potentially reduce stability and fold-to-fold consistency.
2. **Model Assumptions:** Slight deviations in residuals suggest minor violations of linearity or variance assumptions.
3. **Predictor Complexity:** High-dimensional categorical variables introduce sparsity, complicating coefficient interpretation.
4. **Potential Bias and Overfitting:** Some variability in RMSE across folds indicates sensitivity to training data composition.
5. **LightGBM** has three primary weaknesses. 1) its leaf-wise tree growth makes it highly prone to **overfitting**, especially on smaller or noisy datasets; 2) the model is **sensitive to hyper-parameter settings**, meaning performance can vary widely without careful tuning of certain parameters; and 3) LightGBM remains **less interpretable** than traditional the MLR or even classical GBM, making it harder to explain model decisions or justify results in settings requiring transparency.

Impact on Reliability: These factors imply that while predictions are accurate on average, caution is warranted for underrepresented subgroups or outliers. Coefficient interpretation (MLR) should consider potential instability for sparse categories.

8.5 Recommendations for Improvement:

- **GBM:** Conduct **more systematic parameter tuning and repeated cross-validation** to optimize predictive accuracy and stability.
- Use feature engineering to combine sparse levels or create composite indicators to improve interpretability.
- Expand the dataset to increase representation of rare categories and standardize preprocessing to ensure reproducibility.

8.6 Consistency with Prior Work from Table 2

Paper 1 by [Al-Bahrani, A., et al., 2020], resulted in the same rough estimated R^2 and RMSE values on similar LMS datasets albeit for different models. The returned MLR values for this work resulted in a much lower explainability value, 0.49 when compared to 0.62 (Al-Bahrani's results). It has to be recorded that the MLR models used different input variables. Paper 2, by [Zhgang, Y., et al., 2025] additionally accounted for attendance, and quiz scores, etc., **used stratified 10-fold CVs** and the **SHAP analysis** with **strong predictors** reporting: $R^2 \sim 0.89$, $RMSE \sim 4.3\%$, and $AUC=0.953$. Comparatively the same model for this work used different predictors, 5-fold CV and yielded 0.60 explainability. Besides providing a benchmark for the LightGBM model, Zhgang's finding demonstrated a **fairness-aware workflow** (stratified folds, SHAP-based feature inspection) that one can replicate to assess bias and explainability.

Table 2: Sanity-check checklist

Item	CURRENT WORK (LightGBM)	Paper 1 (MLR)	Paper 2 (LightGBM)
Outcome variable	Final course grade (continuous)	Same	Same
Predictors	Time, quiz, assignment not used.	Same (forum, quiz, assignment, time)	Forum, Quiz, assignment, time + demographics

Item	CURRENT WORK (LightGBM)	Paper 1 (MLR)	Paper 2 (LightGBM)
Model	(5-fold CV +_Winsorized)	MLR (k-fold CV)	LightGBM (10-fold CV)
Reported R²	0.60	0.62	0.89
RMSE	6.1%	5.8 %	4.3 %
Interpretability	Coefficient signs + (β values)	Yes (β values)	SHAP values

The researcher is satisfied that irrespective of the models used, that they are able to link observed metrics, diagnostic plots, and fold-wise performance to actionable insights. Together, the summary and critical reflection indicate that the models are effective and generalizable, while providing a roadmap for enhancing reliability and interpretability in future applications. Regarding results from paper 2, **the researcher notes the returned high-powered values in every category and embraces the embedded strategies as future challenges to explore as learning continues.**

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